⭐⭐⭐⭐⭐ 6️⃣ Identify and describe the mathematical aspects of a real-world problem situation

This is the **core competency** assessed.

Students must identify:

- Annual fuel consumption
- Fuel price per unit
- Percentage reduction in fuel use
- Annual cost savings
- Initial investment cost
- Payback time (number of years)

➡️ The task primarily assesses the ability to **extract the mathematical essence from a complex real-world context**.

⭐⭐⭐⭐⭐ 4️⃣ Read, decode and make sense of statements to create a model of the situation

- The problem contains a large amount of information, including:
  - Several pieces of **irrelevant data** (ship length, load capacity, maximum speed)
  - Several **relevant data points** (fuel consumption, price, reduction rate)
- Students must decide:
  - Which information is useful
  - Which information can be ignored

➡️ This strongly assesses **reading comprehension combined with modelling awareness**.

⭐⭐⭐⭐ 2️⃣ Identify the key variables in a model

Key variables include:

- Annual diesel consumption C
- Fuel price per litre p
- Reduction rate $r = 20\%$
- Annual savings S
- Initial investment cost I
- Payback time t

➡️ Correct identification of variables is essential for further analysis.

⭐⭐⭐⭐ 7️⃣ Simplify or decompose a situation to make it mathematically tractable

Students need to decompose the situation into:

1. Annual fuel cost without the kite sail
2. Annual fuel savings
3. Annual monetary savings
4. Cumulative savings compared with the installation cost

➡️ This represents a **typical decomposition process in mathematical modelling**.

⭐⭐⭐⭐ 9️⃣ Translate a problem into a standard mathematical representation or algorithm

For example:

$$\text{Annual fuel cost} = 3{,}500{,}000 \times 0.42$$

$$\text{Annual savings} = 20\% \times \text{Annual fuel cost}$$

$$\text{Payback time} = \frac{2{,}500{,}000}{\text{Annual savings}}$$

➡️ This clearly demonstrates **translation from verbal description to a mathematical algorithm**.

⭐⭐⭐ 1️⃣ Select a mathematical description or representation that describes a problem

- Students implicitly choose:
  - A linear cost model
  - An assumption of constant annual savings
- No graphical representation is required, but proportional reasoning is used

➡️ This is a necessary but low-complexity modelling decision.

⭐⭐⭐ 8️⃣ Recognise aspects corresponding with known mathematical concepts or procedures

- The problem closely resembles familiar concepts such as:
  - Break-even analysis
  - Cost–benefit analysis
  - Percentage reduction
- Recognising these helps students apply known solution strategies efficiently

➡️ This supports efficient problem solving.

⭐⭐⭐ 1 2️⃣ Identify the constraints, assumptions, and simplifications in a mathematical model

Implicit assumptions include:

- Fuel price remains constant
- Annual fuel consumption remains constant
- The 20% reduction applies consistently over time
- Maintenance costs, depreciation, and interest are ignored

➡️ This competency is particularly relevant in higher-quality or extended responses.

⭐⭐ 3️⃣ Select a representation appropriate to the problem context

- Quantities are expressed in:
  - Annual costs
  - Annual savings
  - Number of years
- No visual representations are required

➡️ Present but not a major focus.

⭐⭐ 5️⃣ Recognise mathematical structure in problems or situations

- The structure involves:
  - Linear accumulation of savings
  - A fixed initial cost
- The structure is straightforward and explicit

➡️ A secondary competency.

⭐⭐ 1 0️⃣ / 1 1️⃣ Use mathematical tools (variables, symbols, diagrams)

- Variables and formulas are typically used in a full solution
- However, formal symbolic notation is not explicitly required by the task

➡️ These are **supporting competencies**, not central ones.

⭐⭐⭐⭐⭐ 6️⃣ Identify and describe the mathematical aspects of a real-world problem situation

（识别并描述现实问题中的数学要素与关键变量）

这是本题的**核心能力**：

学生需要识别：

- 年燃油消耗量
- 单位燃油价格
- 节省比例（20%）
- 年节省费用
- 初始投资成本
- 回本时间（年数）

➡️ 本题首先考查的是**“从复杂文字信息中提取数学问题本质”**

⭐⭐⭐⭐⭐ 4️⃣ Read, decode and make sense of statements to create a model of the situation

（阅读、解码文本并构建问题模型）

- 题干信息量大，且包含：
  - 多个无关信息（船长、载重、最大速度）
  - 多个相关信息（油耗、价格、节省比例）
- 学生必须判断：
  - **哪些信息有用**
  - **哪些信息可忽略**

➡️ 强烈考查**文本理解 + 数学建模意识**

⭐⭐⭐⭐ 2️⃣ Identify the key variables in a model

（识别模型中的关键变量）

关键变量包括：

- 年柴油消耗量 C
- 柴油单价 p
- 节省比例 $r = 20\%$
- 年节省费用 S
- 初始成本 I
- 回本年数 t

➡️ 没有清晰变量识别，就无法继续计算

⭐⭐⭐⭐ 7️⃣ Simplify or decompose a situation to make it mathematically tractable

（简化或分解问题）

学生需要将复杂情境拆分为：

1. 年燃油成本（无风筝帆）
2. 年节省燃油量
3. 年节省费用
4. 累计节省 = 安装成本

➡️ 属于**典型的分解式建模过程**

⭐⭐⭐⭐ 9️⃣ Translate a problem into a standard mathematical representation or algorithm

（转化为标准数学表示或算法）

例如：

$$\text{Annual fuel cost} = 3{,}500{,}000 \times 0.42$$

$$\text{Annual savings} = 20\% \times \text{Annual fuel cost}$$

$$\text{Payback time} = \frac{2{,}500{,}000}{\text{Annual savings}}$$

➡️ 明确体现**从文字 → 算法模型**的转化

⭐⭐⭐ 1️⃣ Select a mathematical description or representation that describes a problem

（选择合适的数学描述）

- 学生隐含选择的是：
  - 线性成本模型
  - 年度固定节省假设
- 不涉及函数图像，但涉及**比例与线性关系**

➡️ 属于低复杂度但必要的建模选择

⭐⭐⭐ 8️⃣ Recognise aspects corresponding with known mathematical concepts or procedures

（识别熟悉的问题类型或数学概念）

- 与以下熟悉问题高度一致：
  - 回本周期（break-even analysis）
  - 成本—收益分析
  - 百分比减少

➡️ 有助于学生快速套用成熟思路

⭐⭐⭐ 1 2️⃣ Identify the constraints, assumptions, simplifications in a model

（识别模型中的假设与限制）

隐含假设包括：

- 柴油价格恒定
- 年油耗恒定
- 20% 节省长期有效
- 不考虑维护成本、折旧、利息等

➡️ 在高质量解答或拓展讨论中尤为重要

⭐⭐ 3️⃣ Select a representation appropriate to the problem context

（选择与情境相符的表示方式）

- 使用：
  - 年度成本
  - 年度节省
  - 年数
- 不使用图像，但使用清晰的数量关系

➡️ 有体现，但不是显性考查重点

⭐⭐ 5️⃣ Recognise mathematical structure in problems or situations

（识别问题中的数学结构）

- 结构为：
  - 线性累积节省
  - 固定初始成本
- 但结构相对直接

➡️ 次要能力

⭐⭐ 1 0️⃣ / 1 1️⃣ Use mathematical tools (variables, symbols, diagrams)

（使用符号、变量等数学工具）

- 在完整解答中会使用符号与公式
- 但题目并未强制要求形式化符号表达

➡️ 属于支持性能力